Rohan Devraj

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Education

Georgia Institute of Technology | Atlanta, GA *August 2023 - May 2027*

B.S. in Industrial & Systems Engineering, Minor in Computer Science *GPA: 3.87*

- *Concentrations*: Operations Research, AI & Machine Learning
- *Coursework*: Probability, Adv. Optimization, Data Structures & Algorithms, Stochastics, Databases, AI

Experience

Optum Technology (UnitedHealth Group) Jun. 2025 – Aug. 2025

Data Engineering Intern *Minneapolis, MN*

- Developed a Spark-based Python library deployed in Azure Databricks to compute AHRQ Quality & Patient Safety measures from ICD-10 healthcare data, replacing legacy SAS systems and saving $36K annually.
- Reduced processing time by 90% (10 hrs to less than 1 hr) by optimizing with Pandas UDFs and Databricks.
- Built validation pipelines to benchmark Spark outputs against legacy SAS logic and accelerated health data modernization through fast, in-place cloud processing.

Space Systems Optimization Lab at Georgia Tech Aug. 2024 – Present

Undergraduate Researcher *Atlanta, GA*

- First-author research on multi-scale path planning for planetary rovers, accepted to AIAA ASCEND 2026.
- Developed a large-scale Mixed Integer Linear Program (MILP) optimization model in Python (Pyomo/Gurobi) for rover routing over 5,340 mapped locations using elevation-based energy costs (advised by Dr. Koki Ho).
- Solved model with 52,000+ variables and 62,000+ constraints, achieving optimal route plan in under 8 seconds.
- Designed an adaptive planning framework to handle uncertainty in satellite-derived terrain data, enabling dynamic replanning under partial observability.

Carnegie Mellon Department of Statistics & Data Science Jun. 2024 – Aug. 2024

Summer Undergraduate Research Fellow | stat.cmu.edu/cmsac/sure/2024/showcase/ *Pittsburgh, PA*

- Nationally selected (1 of 18) for a competitive research program in data science applied to healthcare analytics.
- Developed regression and machine learning models (linear, lasso, ridge, decision trees) to analyze the impact of adult health behaviors on child mortality and low birthweight at the county level.
- Evaluated and compared models using 10-fold cross-validation; selected linear regression for its 5% lower root mean squared error (RMSE), enhancing low birthweight prediction accuracy.
- Conducted 10+ analyses on maternal healthcare disparities and applied k-means clustering to identify distinct demographic and behavioral risk groups.

Leadership & Projects

GT Autonomous & Connected Transportation Lab

Undergraduate Research Assistant

- Formulated a multi-objective traffic assignment model in Python using Pyomo and Gurobi to optimize routing across zone-based transportation networks, balancing travel time, congestion, and safety across diverse zones.

Georgia Tech Alumni Association Angel Network

Senior Associate | gtangelnetwork.com

- Lead Georgia Tech's first angel investment network, sourcing 20+ startups from a 100+ deal pipeline and supporting over $6.5M in connected capital.
- Direct meetings with founders and collaborate with investors to craft detailed investment memos, analyzing product, market strategy, financial performance, and growth potential.

Georgia Tech - H. Milton Stewart School of ISyE

Engineering Optimization Tutor

- Tutor 70+ students in optimization topics like linear programming, MILP, solution methods, and network models.

Skills & Activities

Programming/Tools: Python, SQL, Java, R, Excel, PySpark, Azure Databricks, SAS, Git

Libraries & Modeling: Pandas, NumPy, ggplot2, dplyr, Gurobi, Pyomo, GurobiPy, CVXPY

Activities: 180 Degrees Consulting, Aarohi Indian Classical Arts, GT Venture Capital Club, Grand Challenges LLC